

TANISH MISRA

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Machine Intelligence & Security Concentrations), Statistics Minor

August 2025 – December 2028

- Coursework: Problem Solving & OOP, Programming in C, The Data Mine, Linear Algebra, Discrete Mathematics
- Activities: Claude Builder Club, Google Developer Group, Purdue Electric Racing, Electric Vehicle Club, F1 @ Purdue

EXPERIENCE

Toyota Motor North America

Plano, TX

Software Engineering Co-op, Automation Team, Connected Experiences Division (CXD)

May 2026 – August 2026

- Built an AI pipeline converting 1,100+ in-vehicle infotainment (IVI) test cases into automation-ready scripts, cutting authoring time from 3+ hours to 25 seconds per case using Pydantic v2 schema validation.
- Engineered a swappable backend across 3 LLM providers (Azure, Vertex, Ollama) with dynamic few-shot retrieval and a deterministic evaluation framework scoring schema validity, grounding, and reference fidelity.

The Data Mine, Purdue University

West Lafayette, IN

Undergraduate Data Science Researcher (12 completed projects; 2 highlighted below)

January 2026 – Present

- Benchmarked pandas vs Polars on 500K+ row flight datasets, measuring read performance at scale and comparing explicit vs implicit API design patterns for filtering, selection, and type coercion.
- Built a color-coded US state map with GeoPandas and Folium, ingesting Census boundary data, reprojecting coordinate systems, normalizing state names for a spatial join, and rendering styled interactive maps.

Q Spark Group (QSG)

Remote

Data, Analytics, and AI Intern

March 2024 – July 2025

- Led a PII privacy initiative training Transformer NER models on 1M+ synthetic records (recall to 90%) and analyzed 250,000+ SQL records for metadata quality checks to organize company-wide data access.

Purdue Electric Racing

West Lafayette, IN

Firmware Developer, Software Team

August 2025 – Present

- Developed embedded C firmware validating git-hash consistency across 7 STM32 node types (90 CAN messages) at boot, classifying mismatches into 3 severity levels so failures trace to the exact version without manual repo inspection.

Electric Vehicle Club

West Lafayette, IN

Software Developer, Autonomous Team

August 2025 – Present

- Co-developed an OpenCV vision node within a 9-node ROS2 system processing 60 Hz camera frames to detect lane edges via color segmentation, track edges across frames, and compute steering angles for an autonomous kart.

PROJECTS

Telogify (F1 Telemetry Intelligence) | *LangGraph, PostgreSQL, Anthropic API, React* August 2026

- Built an F1 analysis platform that ingests full weekends via FastF1, precomputes every corner speed, stint degradation, and sector split into Postgres, then generates insights through a LangGraph ReAct agent bound to 12+ query tools so all numbers come from deterministic reads, not the model's training data.
- Prevented hallucination with a blacklist guardrail rejecting unsupported claims and a validation pass cross-checking every cited number against the tool-call trace; deployed at telogify.com with an automated email digest.

Candle for Choroideremia (CHM) | *FastAPI, PostgreSQL, pgvector, OpenAI API* April 2026

- Built a patient-facing research dashboard with automated ingestion of 29 trials and 475 publications, a RAG pipeline (pgvector cosine similarity & reranking, intent classification, query expansion), and medical safety guardrails; adopted at candleforchm.org after user acceptance testing by 3 blind patients.

Securify | *LangGraph, RoBERTa, OpenAI API* | *AlgoFest 2026 Hackathon Semi-Finalist* April 2026

- Architected a 4-node LangGraph pipeline (router, retrieval, synthesis, security) that redacts PII via a fine-tuned RoBERTa NER model (160K examples, 10 entity classes, 0.940 OOD micro F1), blocks prompt injections with 18 regex filters, and scans output for PII leaks; tested across 3 document types (medical, financial, HR).

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, SQL

AI/ML & Data: LangChain, LangGraph, HuggingFace, Anthropic API, OpenAI API, PyTorch, OpenCV, PostgreSQL

Developer Tools: Git/GitHub, Docker, Codex, Claude Code, REST APIs, React, Vercel, Railway